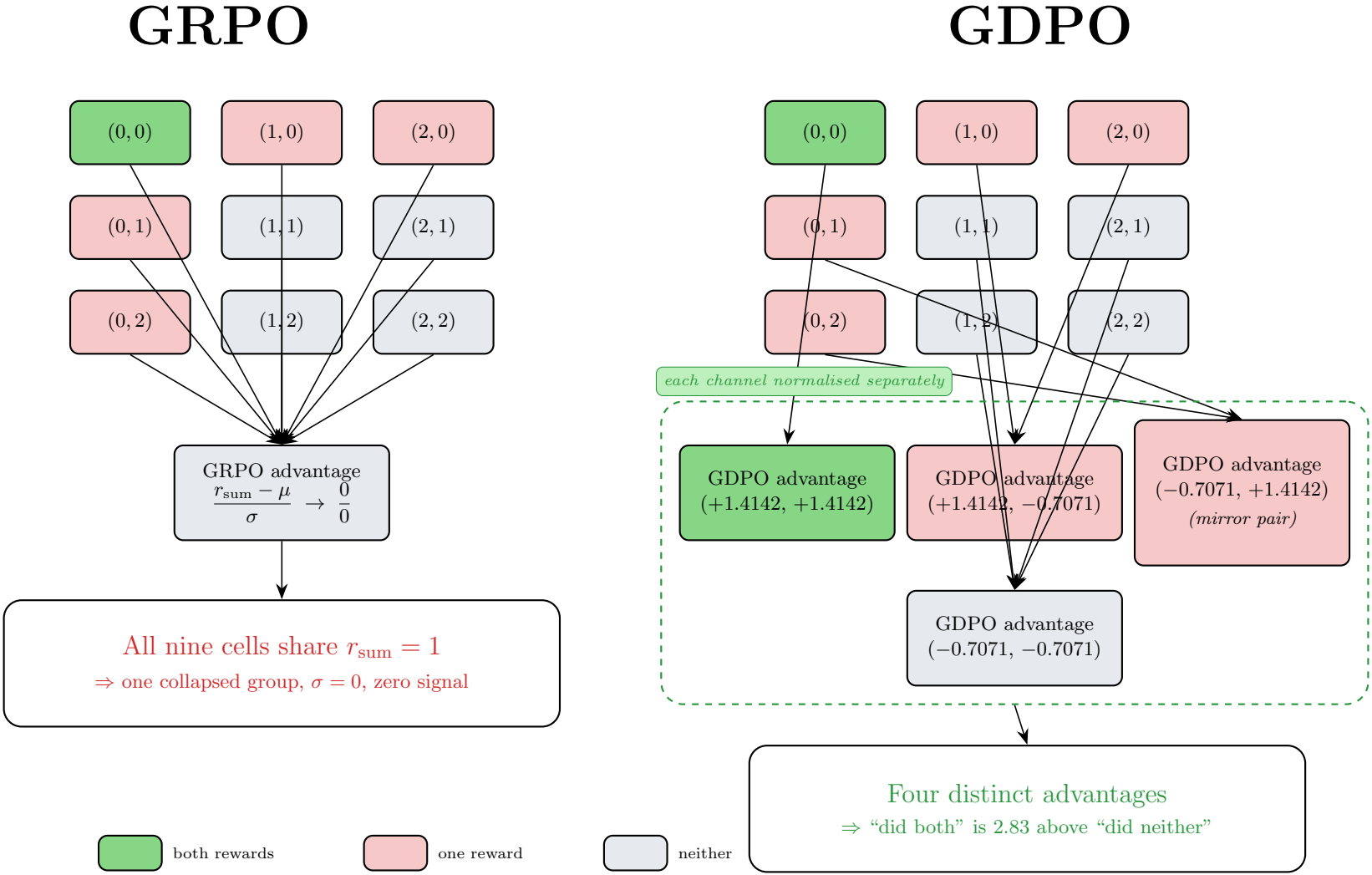

GRPO
(0, 0)
(1, 0)
(2, 0)
(0, 1)
(1, 1)
(2, 1)
(0, 2)
(1, 2)
(2, 2)
GRPO advantage
$\frac{r_{\text{sum}} - \mu}{\sigma} \rightarrow \frac{0}{0}$
All nine cells share $r_{\text{sum}} = 1$
⇒ one collapsed group, $\sigma = 0$, zero signal
GDPO
each channel normalised separately
GDPO advantage
(+1.4142, +1.4142)
GDPO advantage
(+1.4142, −0.7071)
GDPO advantage
(−0.7071, +1.4142)
(mirror pair)
GDPO advantage
(−0.7071, −0.7071)
Four distinct advantages
⇒ "did both" is 2.83 above "did neither"
both rewards
one reward
neither